

Building Neural Networks

General Project Report

Current status: Versions 1-4 implemented and evaluated; Version 5 is planned.

1. Project Overview

Building Neural Networks is a compact educational project for reconstructing neural networks from scratch with NumPy. The goal is to expose weights, biases, activations, losses, gradients, backpropagation, and training loops that are normally hidden by high-level frameworks.

2. Development Principle

One version, one major concept. Each version keeps the previous logic and adds the minimum necessary to introduce the next idea.

3. Dataset and Evaluation

The project uses Fashion-MNIST: 28 x 28 grayscale clothing images represented by 784 normalized pixel values. Version 1 uses only T-shirt/top and Trouser. Versions 2-4 use all 10 classes and are evaluated on the full provided training and test sets.

4. Version 4

Version 4 keeps the 784 -> 64 -> 10 network from Version 3 and changes the training strategy: the data are shuffled each epoch, split into mini-batches of 64, and the weights are updated after every batch. The reported training loss is the mean batch loss of the final epoch. np.clip is used inside the logarithm for numerical stability.

5. Roadmap

Version	Name	Architecture	Status / main task
V1	Simple Single Neuron	784 -> 1	Implemented - binary classification
V2	Multiclass Neural Network	784 -> 10	Implemented - 10-class softmax
V3	Single Hidden Layer Network	784 -> 64 -> 10	Implemented - hidden layer + backprop
V4	Mini-Batch Neural Network	784 -> 64 -> 10	Implemented - mini-batch + shuffle
V5	Simple Deep Neural Network	784 -> 128 -> 64 -> 10	Planned - deeper network

6. Results

Metric	V1	V2	V3	V4
Train loss	0.0305	0.5092	0.5895	0.0695
Train accuracy	98.97%	83.13%	79.03%	95.75%
Test accuracy	99.15%	83.42%	79.02%	87.28%

The versions are educational steps, so their accuracies are not directly comparable as model-quality rankings: each version changes a different concept and V1 uses only two classes.